

Taran Govindu

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EDUCATION

Georgia Institute of Technology

Expected Graduation May 2028

Bachelor of Science in Aerospace Engineering, Minor in Computer Science

4.0 GPA

- Relevant Coursework: Object-Oriented Programming, Data Structures and Algorithms, Dynamics, Circuits and Electronics, Thermodynamics and Fluids, MATLAB, CAD

SKILLS

Software & Tools: SolidWorks, MAVLink, Git, Linux (Ubuntu), Soldering, 3D Printing, Microsoft Office

Programming Languages: Python, Rust, C++, MATLAB, Java

Controls & Estimation: LQR/Optimal Control, Kalman Filtering (EKF), Reinforcement Learning (RL), Visual-Inertial Odometry (VIO), 6-DOF Rigid-Body Modeling, PnP Pose Estimation

ACHIEVEMENTS

Eagle Scout, 2024

Startup Co-Founder – raised \$155,000 in pre-seed funding at a \$1.5M valuation building a Python-based AI system for GPR bridge inspection

EXPERIENCE

Intelligent Cyber-Physical Systems (ICPS) Laboratory at Georgia Tech

Atlanta, GA

Researcher

Jun 2026 – Present

- Reproduced a Python control framework that switches a model-based Linear-Quadratic Regulator (LQR) to a reinforcement-learning (RL)-augmented controller once real-time mismatch with the true nonlinear plant crosses an optimality-based threshold
- Extended the framework in Python to a rocket pitch-attitude example across three tiers of increasing complexity: aerodynamic nonlinearity, mass-burn dynamics, and propellant slosh, and verified the learned control gains converged to the analytic optimal solution within 5 percent
- Porting the same switching framework to a simulated quadrotor, deriving a full nonlinear hover model and an LQR controller directly from the vehicle's measured motor and inertia properties to prepare for hardware deployment

Propulsive Landers Rocket Team at Georgia Tech

Atlanta, GA

GNC Engineer

Jan 2026 – Present

- Modeled high-fidelity aerodynamics with a Gauss-Markov gust model and coefficient lookup tables in an existing 6-DOF Rust simulation, a testbed for GNC algorithms on the team's propulsive-landing rocket
- Integrated valve actuation dynamics and pressure transducer response as propulsion feed system models and wired the reaction control system (RCS) valves into roll-axis control

Exascale CFD VIP at Georgia Tech

Atlanta, GA

Researcher

Jan 2026 – Jun 2026

- Extended an existing 1D Godunov scheme in C++ to a full 2D solver for the Burgers' equation with built-in conservation and stability checks, generating benchmark data for our machine-learning surrogate models
- Implemented canonical test cases with known solutions (Gaussian pulse, N-wave, and others) and a tool to compare how the Godunov and Lax-Wendroff schemes smooth out sharp features across viscosity levels

PROJECTS

Anduril AI Grand Prix | Autonomous Drone Racing Competition

May 2026 – Aug 2026

- Built a vision-only perception and state-estimation stack in Python for an autonomous drone on simulator that allowed only a camera and IMU, detecting gates with a Perspective-n-Point (PnP) pose solver
- Fused gate detections with inertial data in a Python six-state Extended Kalman Filter (EKF) for visual-inertial odometry (VIO), cutting position error 6.5x versus dead-reckoning and velocity error 2.8x versus vision-only estimates
- Built the offline accuracy-scoring framework that benchmarked perception and state-estimation error against ground-truth odometry by range and surfaced a systematic vertical bias in the detector

Acoustic Drone Detection System

Apr 2026

- Reverse-engineered a speaker to function as a microphone and designed and 3D-printed a parabolic reflector in SolidWorks to focus incoming sound onto the microphone, then soldered the sensors to LED indicators and a digital display, building a full system for under \$25
- Performed live Fast Fourier Transforms on an ESP32 microcontroller to isolate drone propeller frequency bands from background noise, then trained a neural network on 20,000+ audio samples in Python and deployed the model in C++ for on-device inference at 85% detection accuracy